

Half wave rectification

1. Average Value (V_avg)

$$V_{avg} = \int_0^T \frac{V}{T} dt$$

$$V_{avg} = \left(\frac{1}{T}\right) \int_0^T V_m \sin \omega t dt$$

$$V_{avg} = \left(\frac{V_m}{T}\right) \int_0^T \sin \omega t dt$$

$$V_{avg} = \left(\frac{V_m}{T}\right) \left[\frac{-\cos \omega t}{\omega} \right]_0^T \quad \left[\int \sin ax dt = \frac{-\cos at}{a} \right]$$

$$V_{avg} = -\left(\frac{V_m}{\omega T}\right) [\cos \omega t]_0^T$$

$$V_{avg} = -\left(\frac{V_m}{2\pi}\right) \left[\cos \omega \frac{T}{2} - \cos \omega 0 \right]$$

$$V_{avg} = -\left(\frac{V_m}{2\pi}\right) \left[\cos \frac{2\pi}{2} - \cos \omega 0 \right]$$

$$V_{avg} = -\left(\frac{V_m}{2\pi}\right) [\cos \pi - \cos \omega 0]$$

$$V_{avg} = -\left(\frac{V_m}{2\pi}\right) [-1 - 1]$$

$$V_{avg} = -\left(\frac{V_m}{2\pi}\right) [-2]$$

$$V_{avg} = \left(\frac{V_m}{\pi}\right)$$

$$V_{avg} = (0.318 V_m)$$

2. RMS Value (V_{rms})

$$V_{rms} = \sqrt{\overline{V}}$$

$$V_{rms}^2 = \frac{V^2}{T}$$

$$V_{rms}^2 = \int_0^T \frac{V^2}{T} dt$$

$$V_{rms}^2 = \left(\frac{1}{T}\right) \int_0^T (V_m \sin \omega t)^2 dt$$

$$V_{rms}^2 = \left(\frac{V_m^2}{T}\right) \int_0^T \sin^2 \omega t dt$$

$$V_{rms}^2 = \left(\frac{V_m^2}{T}\right) \int_0^T \frac{2}{2} \sin^2 \omega t dt$$

$$V_{rms}^2 = \left(\frac{V_m^2}{2T}\right) \int_0^T [1 - \cos 2\omega t] dt$$

$$V_{rms}^2 = \left(\frac{V_m^2}{2T}\right) \int_0^T [1 - \cos 2\omega t] dt$$

$$V_{rms}^2 = \left(\frac{V_m^2}{2T}\right) \left[t - \frac{\sin 2\omega t}{2\omega} \right]_0^T \quad \left[\int \cos ax dt = \frac{\sin ax}{a} \right]$$

$$V_{rms}^2 = \left(\frac{V_m^2}{2T}\right) \left[\left(\frac{T}{2} - 0 \right) - \frac{1}{2\omega} \left(\sin 2\omega \frac{T}{2} - \sin 2\omega 0 \right) \right]$$

$$V_{rms}^2 = \left(\frac{V_m^2}{2T}\right) \left[\frac{T}{2} - \frac{1}{2\omega} (\sin 2\pi - \sin 0) \right]$$

$$V_{rms}^2 = \left(\frac{V_m^2}{2T}\right) \left[\frac{T}{2} \right]$$

$$V_{rms}^2 = \left(\frac{V_m^2}{4}\right)$$

$$V_{rms} = \left(\frac{V_m}{2}\right)$$

$$V_{rms} = (0.5 V_m)$$

Full wave rectification

1. Average Value (V_{avg})

$$V_{avg} = \int_0^{\frac{T}{2}} \frac{V}{\frac{T}{2}} dt$$

$$V_{avg} = \left(\frac{1}{\frac{T}{2}}\right) \int_0^{\frac{T}{2}} V_m \sin \omega t dt$$

$$V_{avg} = \left(\frac{2V_m}{T}\right) \int_0^{\frac{T}{2}} \sin \omega t dt$$

$$V_{avg} = \left(\frac{2V_m}{T}\right) \left[\frac{-\cos \omega t}{\omega} \right]_0^{\frac{T}{2}} \quad \left[\int \sin ax dt = \frac{-\cos at}{a} \right]$$

$$V_{avg} = -\left(\frac{2V_m}{\omega T}\right) [\cos \omega t]_0^{\frac{T}{2}}$$

$$V_{avg} = -\left(\frac{2V_m}{2\pi}\right) \left[\cos \omega \frac{T}{2} - \cos \omega 0 \right]$$

$$V_{avg} = -\left(\frac{2V_m}{2\pi}\right) \left[\cos \frac{2\pi}{2} - \cos \omega 0 \right]$$

$$V_{avg} = -\left(\frac{V_m}{\pi}\right) [\cos \pi - \cos \omega 0]$$

$$V_{avg} = -\left(\frac{V_m}{\pi}\right) [-1 - 1]$$

$$V_{avg} = -\left(\frac{V_m}{\pi}\right) [-2]$$

$$V_{avg} = \left(\frac{2V_m}{\pi}\right)$$

$$V_{avg} = (0.636 V_m)$$

$$V_{rms} = \frac{V}{\sqrt{2}}$$

$$\Rightarrow V_{rms}^2 = \frac{V^2}{2}$$

$$\Rightarrow V_{rms}^2 = \frac{2}{T} \int_0^{\frac{T}{2}} V_m^2 \sin^2 \omega t \, dt$$

$$\Rightarrow V_{rms}^2 = \frac{2V_m^2}{T} \int_0^{\frac{T}{2}} \sin^2 \omega t \, dt$$

$$\Rightarrow V_{rms}^2 = \frac{2V_m^2}{T} \int_0^{\frac{T}{2}} \frac{1 - \cos 2\omega t}{2} \, dt$$

$$\Rightarrow V_{rms}^2 = \frac{2V_m^2}{T} \int_0^{\frac{T}{2}} \frac{1}{2} (1 - \cos 2\omega t) \, dt$$

$$\Rightarrow V_{rms}^2 = \frac{V_m^2}{T} \int_0^{\frac{T}{2}} (1 - \cos 2\omega t) \, dt$$

$$\Rightarrow V_{rms}^2 = \frac{V_m^2}{T} \left[t - \frac{\sin 2\omega t}{2\omega} \right]_0^{\frac{T}{2}}$$

$$\Rightarrow V_{rms}^2 = \frac{V_m^2}{T} \left[\left(\frac{T}{2} - 0 \right) - \frac{1}{2\omega} (\sin 2\omega \frac{T}{2} - \sin 2\omega 0) \right]$$

$$\Rightarrow V_{rms}^2 = \frac{V_m^2}{T} \left[\frac{T}{2} - \frac{1}{2\omega} (\sin 2\pi - 0) \right]$$

$$\Rightarrow V_{rms}^2 = \frac{V_m^2}{T} \left[\frac{T}{2} - 0 \right]$$

2)

$$V_{rms}^2 = \frac{V_m^2}{T} \cdot \frac{T}{2} = \frac{V_m^2}{2}$$

$$\Rightarrow V_{rms} = \frac{V_m}{\sqrt{2}} = 0.707 V_m$$